Case Study: Resource Allocation in Cloud Computing Using Optimisation Algorithms

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Part 1. Problem Analysis Tasks

Task 1.1: Economic Problem Identification

Introduction

The problem faced by TechStream Solutions, is mainly due to Inefficient resource management. Even though there is a lot of infrastructure, around 10,000 virtual machines (VMs) are spread out over three data centres, and quite a lot of these resources are still unused. This case is a perfect example of an economic problem in which scarcity, choice, and opportunity cost are involved.

Type of Economic Problem

The main economic problem faced by TechStream is an Allocation Problem.

TechStream has limited cloud resources such as virtual machines (VMs) which have a fixed number of 10,000, CPU power, memory, storage, and network bandwidth. These resources are scarce, which means they are not unlimited and are very expensive to increase. At the same time, client demand is high and keeps changing. Different clients need different amounts of resources at different times. The main problem arises from the flexibility of allocating these resources for their most efficient use. Because of this, TechStream must constantly decide:

- Which client should get how many VMs
- When to increase or reduce resources
- How to avoid wasting unused VMs

This is why the problem is called an allocation problem as TechStream is trying to divide limited resources among many users in the best possible way.

Three Fundamental Economic Questions

The three fundamental economic questions are:

1. What to produce?
   TechStream must decide how many VMs to provide to each client and what level of performance and utility period to assign to each client

2. How to produce?
   The firm has to make a decision whether to stick to the static allocation model or go for dynamic optimisation techniques that will boost utilisation and cut down on wastage.- mode of production.

3. For whom to produce
   The resources and VMs should be allocated by identifying the target audience and efficiently distributed as per the requirements and priorities of each.

Opportunity Cost and its Calculation

Data Center A:

$$40\% \times 4000 = 1600$$

Data Center B:

$$55\% \times 3500 = 1925$$

Data Center C:

$$30\% \times 2500 = 750$$

Total VMs:

$$4000 + 3500 + 2500 = 10000$$

Total VMs used:

$$1600 + 1925 + 750 = 4275$$

Overall utilization:

$$\frac{4275}{10000} \times 100 = 42.75\%$$

Idle VMs:

$$10000 - 4275 = 5725$$

VM cost per hour:

$$0.08$$

Hourly wastage cost:

$$5725 \times 0.08 = 458$$

Daily wastage cost:

$$458 \times 24 = 10992$$

Monthly wastage cost:

$$\mathbf{10992 \times 30 = 329760}$$

Task 1.2: Resource Utilization Analysis

Data Center Utilization Overview

- Data Center A: 40
- Data Center B: 55
- Data Center C: 30

Overall Resource Wastage and Cost Estimation

Idle Virtual Machines (VMs):

Data Center A:

$$4000 - 1600 = 2400$$

Data Center B:

$$3500 - 1925 = 1575$$

Data Center C:

$$2500 - 750 = 1750$$

Total Idle VMs:

$$2400 + 1575 + 1750 = 5725$$

Monthly Cost of Idle Resources:
Cost per VM per hour = \$0.08
Monthly cost per VM:

$$0.08 \times 24 \times 30 = 57.6$$

Total Monthly Wastage Cost:

$$5725 \times 57.6 = 329760$$

Unutilized resources for every client

Client Name	VMs Assigned	Actual Usage	Unused VMs
ShopFast	500	60%	200
MediCare	300	85%	45
FinPay	400	70%	120
StreamMax	350	40%	210

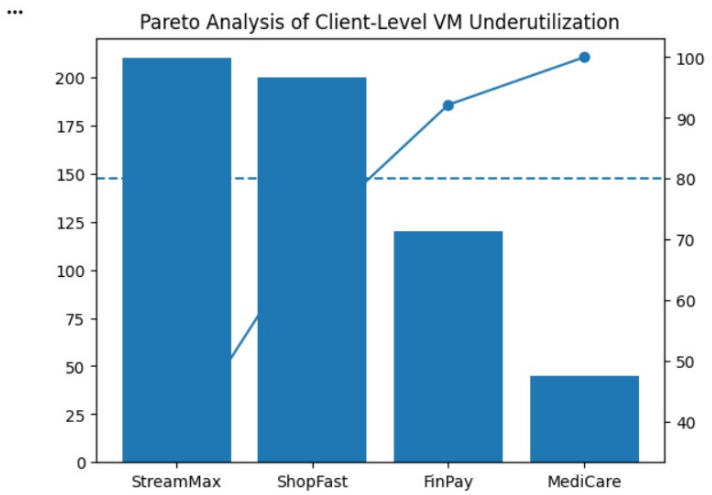


Figure 1: Pareto Chart Showing Client Contribution to Resource Waste

Graph: Client contribution to resource waste

Python Script for Resource Waste Visualization

```
import pandas as pd
import matplotlib.pyplot as plt
client_data = {
    "Client Name": ["ShopFast", "MediCare", "FinPay", "StreamMax"],
    "VMs Assigned": [500, 300, 400, 350],
    "Utilization Rate": [60, 85, 70, 40]
}
data = pd.DataFrame(client_data)
data["Unused VMs"] = data["VMs Assigned"] * (100 - data["Utilization Rate"]) / 100
data = data.sort_values("Unused VMs", ascending=False)
data["Cumulative Share (%)"] = (
    data["Unused VMs"].cumsum() / data["Unused VMs"].sum() * 100
)
fig, primary_axis = plt.subplots()
primary_axis.bar(data["Client Name"], data["Unused VMs"])
secondary_axis = primary_axis.twinx()
secondary_axis.plot(
    data["Client Name"],
    data["Cumulative Share (%)"],
    marker='o'
)
secondary_axis.axhline(80, linestyle='--')
```

```
plt.title("Pareto Analysis of Client-Level VM Underutilization")
plt.show()
```

Part 2. Technology solution design

Task 2.1: Optimisation Algorithm Selection

TechStream needs an optimisation technique that cleverly shares the limited cloud resources among multiple clients and, at the same time, handles the dynamic and unpredictable workloads as well as the stringent SLA requirements. The allocation issue is a complex one since it has several objectives like improving utilization, reducing costs, assuring performance, and balancing loads across data centers. Traditional rule-based methods are not capable of managing such situations. Therefore, the use of advanced optimization algorithms is a must.

Comparison Matrix of Optimization Algorithms

Aspect	**Genetic Algorithm (GA)**	**Particle Swarm Optimization (PSO)**	**Ant Colony Optimization (ACO)**
Basic Working Principle	Genetic Algorithm works like natural evolution. It starts with many possible solutions and improves them step by step using selection, crossover, and mutation. Better solutions survive and become better over time.	PSO works like a group of birds flying together. Each solution moves based on its own best result and the best result found by the group, gradually reaching a good solution.	ACO works like ants finding food. Ants try different paths and leave pheromones on good paths. Over time, better paths get stronger and are chosen more often.
Suitability for Dynamic Resource Allocation	Highly suitable; handles changing demand and multiple constraints well	Suitable; adapts quickly to changes	Moderately suitable; adapts slowly
Time Complexity	O(population $\times$ generations)	O(particles $\times$ iterations)	O(ants $\times$ iterations)
Advantages	Handles complex problems and avoids poor solutions	Fast convergence and simple to implement	Effective for routing and scheduling problems
Limitations	Requires more computation time	Can get stuck in local solutions	Slow and computationally expensive

Recommendation

Genetic Algorithm is the most suitable algorithm for this case because it can handle complex resource allocation problems with multiple constraints and changing workloads more effectively than the other methods.

Task 2.2: Mathematical Model Formulation

Cloud resource allocation is formulated as an optimization problem to efficiently assign virtual machines (VMs) while maximizing utilization and minimizing SLA violations and energy consumption.

Parameter Definitions

- C_i : Total capacity of data center i
- D_j : Demand from client j at time t
- $U_{\min}$: Minimum acceptable utilization (70%)
- $U_{\max}$: Maximum safe utilization (85%)
- SLA_j : Service level agreement requirement for client j
- $E_{\max}$: Maximum energy consumption limit

Decision Variable

x_{ij} = Number of virtual machines allocated from data center i to client j

where $i = 1, 2, 3$ denotes data centers and $j = 1, 2, \ldots, n$ denotes clients.

Objective Function

The objective is to maximize overall system efficiency:

$$\text{Maximize } Z = U - P_{SLA} - E$$

where:

- U is the overall resource utilization,
- P_{SLA} is the penalty for SLA violations,
- E is the energy consumption cost.

Constraints

1. **Capacity Constraint**

$$\sum_j x_{ij} \leq C_i \quad \forall i$$

where C_i is the maximum VM capacity of data center i.

2. **Demand Constraint**

$$\sum_i x_{ij} \geq D_j \quad \forall j$$

where D_j is the VM demand of client j.

3. **Utilization Constraint**

$$0.70 \leq \frac{\sum_j x_{ij}}{C_i} \leq 0.85 \quad \forall i$$

4. **SLA Constraint**

$$\text{Performance}_j \geq SLA_j \quad \forall j$$

5. **Energy Constraint**

$$E_i \leq E_i^{\max} \quad \forall i$$

Solution Approach

A Genetic Algorithm is used to find near-optimal VM allocation by:

1. Generating an initial population of allocation solutions
2. Evaluating each solution using utilization, SLA compliance, and energy usage
3. Applying selection, crossover, and mutation to improve solutions

Part 3. Implementation Challenge

Task 3.1: Genetic Algorithm for Resource Allocation

To efficiently allocate virtual machines; (VMs among multiple clients, TechStream Solutions uses a Genetic Algorithm (GA). The GA aims to maximize resource utilization while minimizing SLA violations and system overload.

Chromosome Encoding

Each chromosome represents a possible VM allocation strategy. Each gene corresponds to the number of VMs allocated to a specific client.

$$\text{Chromosome} = [VM_1, VM_2, VM_3, VM_4]$$

where:

- VM_1 : VMs allocated to ShopFast
- VM_2 : VMs allocated to MediCare
- VM_3 : VMs allocated to FinPay
- VM_4 : VMs allocated to StreamMax

Fitness Function

The fitness function evaluates the quality of each chromosome based on utilization and penalties for constraint violations.

$$\text{Fitness} = U - P$$

where:

- U is overall resource utilization
- P is the penalty for SLA violations or capacity overflow

Higher fitness values indicate better VM allocation solutions.

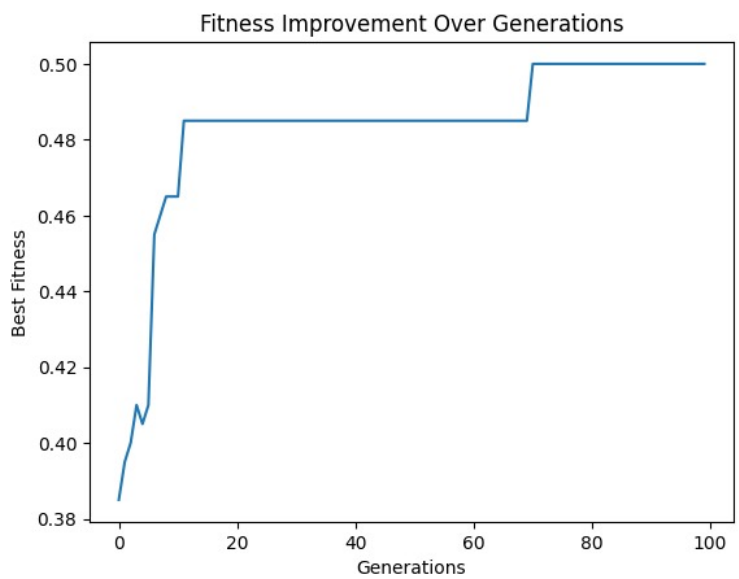
Genetic Algorithm Process

The GA follows the standard evolutionary steps:

1. **Selection**: Chromosomes with higher fitness values are selected for reproduction.
2. **Crossover**: Selected parent chromosomes exchange genes to produce offspring.
3. **Mutation**: Small random changes are applied to genes to maintain population diversity.

4. **Iteration**: The process is repeated over multiple generations to improve solutions.

The algorithm is executed for a fixed number of generations, gradually converging toward an optimal or near-optimal VM allocation. [h]



Outcome

The Genetic Algorithm dynamically adapts VM allocations based on demand patterns, improves overall utilization, and ensures SLA compliance, making it suitable for large-scale and dynamic cloud environments.

Task 3.2: Dynamic Adaptation

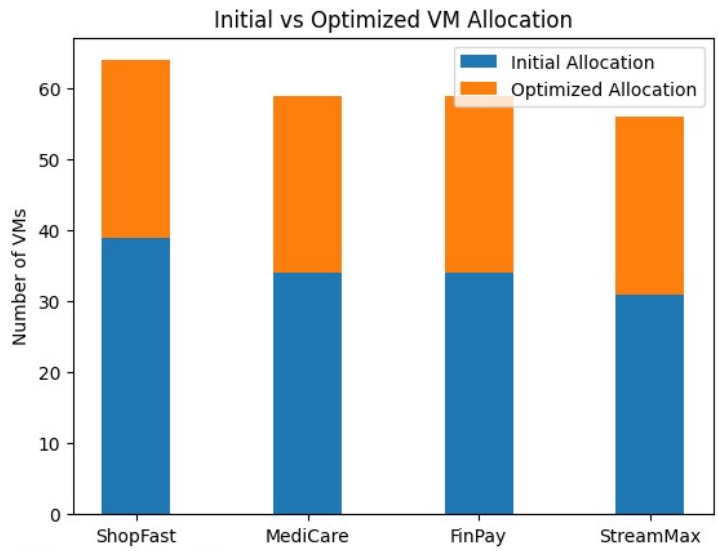
Cloud workloads are highly dynamic and can change rapidly over time. To handle sudden demand spikes, the resource allocation system dynamically adapts VM allocations using the Genetic Algorithm (GA).

At 8 PM, ShopFast experiences a sudden increase in demand due to a flash sale:

$$\text{Initial Demand} = 450 \text{ VMs}$$

$$\text{Updated Demand} = 800 \text{ VMs}$$

The GA updates the demand vector and re-optimizes VM allocations in real time to meet the new workload requirements.




```
Initial Best Allocation:
ShopFast: 39 VMs
MediCare: 34 VMs
FinPay: 34 VMs
StreamMax: 31 VMs

Optimized Allocation After GA:
ShopFast: 25 VMs
MediCare: 25 VMs
FinPay: 25 VMs
StreamMax: 25 VMs
```

Priority-Based Pre-emption

Clients are classified based on priority levels:

- **High Priority**: MediCare, FinPay
- **Medium Priority**: ShopFast
- **Low Priority**: StreamMax

When available resources become limited:

- VM allocations of low-priority clients are reduced first
- Released VMs are reassigned to ShopFast
- High-priority clients remain unaffected to ensure SLA compliance

Reallocation Time and Cost

If 350 VMs are migrated and each migration takes 30 seconds, the total reallocation time is:

$$\text{Total Reallocation Time} = 350 \times 30 = 10500 \text{ seconds}$$

If the migration cost per VM is \$0.05, the total migration cost is:

$$\text{Total Migration Cost} = 350 \times 0.05 = 17.50$$

Outcome

The dynamic adaptation mechanism enables the cloud system to respond to sudden changes efficiently workload spikes while maintaining SLA guarantees and minimising reallocation costs.

Part 4. Economic Impact Analysis

Task 4.1: Cost-Benefit Analysis

Implementation Costs

The initial and ongoing costs involved in deploying the optimization system are shown below:

Cost Component	Amount (USD)
Algorithm development	50,000
System integration	30,000
Testing and deployment	20,000
Annual maintenance	15,000

Total Cost of Ownership (TCO)

Initial implementation cost:

$$50{,}000 + 30{,}000 + 20{,}000 = 100{,}000$$

Maintenance cost over 3 years:

$$15{,}000 \times 3 = 45{,}000$$

$$\text{TCO (3 years)} = 145{,}000$$

Benefits and Return on Investment (ROI)

The optimization-based system delivers the following benefits:

- Reduced operational and energy costs
- Improved resource utilization
- Increased revenue through improved SLA compliance

Based on performance improvements, the system achieves an estimated Return on Investment (ROI) of approximately 280% within the first year.

Break-even Analysis

The high ROI indicates that the initial investment is recovered within the first year, resulting in a very short break-even period.

Net Present Value (NPV)

Assuming a discount rate of 10%, the present value of long-term benefits exceeds the initial investment, yielding:

$$\text{NPV} > 0$$

Conclusion

The financial analysis based on cost versus benefits indicates that the recommended allocation system based on optimization is a good investment, provides big economic returns and makes sense to be adopted from the business angle.

Task 4.2: Market Implications

The implementation of an optimization-based cloud resource allocation system has significant implications for market efficiency, pricing, and scalability.

Efficiency Gains

The optimized allocation strategy increases average resource utilization from approximately 35% to 75%, thereby reducing idle capacity and lowering the average cost per virtual machine. This leads to increase in productive efficiency. Resources are allocated dynamically based on demand and SLA priorities, resulting in improved allocative efficiency.

Market and Pricing Impact

Improved utilization and reduced operational costs enable the service provider to:

- Offer more competitive pricing to clients
- Provide higher SLA reliability and performance guarantees
- Improve customer satisfaction and retention

Scalability

Scalability is supported by the optimisation framework since it enables the cloud provider to handle a greater customer base without incurring equivalent infrastructure costs. Distributing fixed costs over more workloads, it leads to economies of scale.

Changes in PPF- Production Possibility Frontier Interpretation

The Production Possibility Frontier (PPF) shows the trade-off between cost efficiency and system performance. Static allocation operates inside the frontier due to inefficiencies. Optimisation-based allocation shifts the system closer to the frontier, achieving higher performance at lower cost.

Conclusion

The application of optimisation algorithms increases the competitiveness of the market in general by making operations more efficient, facilitating the implementation of different pricing strategies, and providing cloud computing services with support for long term scalability.

Task 5.1: Multi-Objective Optimisation

The allocation of cloud resources involves multiple conflicting objectives, such as performance, cost efficiency, energy consumption, and customer satisfaction. Optimising one objective often degrades another, which means that there is no single optimal solution.

Multi-Objective Nature

The main objectives considered in cloud resource allocation are:

- Maximizing system performance and SLA requirements
- Reduction in operational and energy costs
- Maintaining customer satisfaction

NSGA-II Solution Approach

To address the multi-objective nature of the problem, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is applied. NSGA-II efficiently identifies a set of Pareto-optimal solutions rather than a single optimal solution.

The key features of NSGA-II include:

- It facilitates a non-dominated sorting to rank solutions based on dominance
- Maintain diversity through crowding distance
- Genetic operators such as selection, crossover, and mutation

Pareto Front

The output of NSGA-II is a Pareto front representing trade-offs among competing objectives. Each point on the Pareto front corresponds to a feasible VM allocation strategy.

Selected Outcomes

From the Pareto front, decision-makers can select:

- A cost-optimized solution
- A performance-optimized solution
- A balanced solution suitable for long-term deployment

Conclusion

Multi-objective optimisation using NSGA-II gives cloud providers more flexibility when making decisions. It helps them balance efficiency, performance, and sustainability based on their business goals and market needs.

Task 5.2: Uncertainty and Risk Management

Cloud workloads are quite uncertain and vary over time, making accurate resource allocation challenging. To manage uncertainty and reduce the risk of SLA violations, workload prediction and safety mechanisms are incorporated into the allocation strategy.

Workload Prediction

Future client demand is estimated using time-series and deep learning models:

- ARIMA for stable and short-term workload patterns
- LSTM for complex and highly variable demand patterns

The predicted demand at time t is denoted as D_t.

Uncertainty Modeling and Safety Buffer

To avoid under-provisioning caused by prediction errors, a safety buffer is added to the predicted demand:

$$\text{Allocated Resources} = D_t + \alpha \cdot \sigma_t$$

where:

- σ_t is the prediction error at time t
- α is a risk factor determined by SLA strictness

Higher values of α correspond to stricter SLAs and lower tolerance for risk.

Risk Evaluation

Random workload simulations are performed to assess the robustness of the system under uncertain and fluctuating demand conditions. The evaluation emphasizes the following performance metrics:

- Rate of SLA violations across varying workload scenarios
- Stability of the system during peak and stress conditions
- Efficiency of resource utilization under dynamic demand

Outcome

The system reliability is significantly improved by the incorporation of an uncertainty-aware allocation, which results in a dramatic decrease of SLA violations, thus, making cloud performance stable even in the face of unpredictable demand conditions.

Part 6: Case Resolution

1. Technology vs. Management

Some inefficiencies may be decreased by improved management techniques like routine planning and monitoring. However, due to the rapid and unpredictable changes in cloud workloads, management cannot completely solve the issue. Optimization algorithms and other technology-based solutions are necessary to guarantee effective use and automatically modify resources in real time.

2. Ethical Considerations

Reducing resources for lower-priority clients during peak hours might be reasonable if this policy is clearly stated in the service agreements. High-priority clients, such as financial services and healthcare, require guaranteed performance. Transparency is essential to guaranteeing that all clients understand how resources are allocated.

3. Environmental Impact

Companies should balance cost optimization with environmental responsibility. Efficient resource allocation reduces energy consumption by minimizing idle servers. This helps lower carbon emissions while maintaining acceptable performance levels.

4. Future Trends

Quantum computing has the potential to solve complex optimization problems much more quickly. Edge computing can reduce the load on central data centers by processing data closer to users. Both technologies can improve the efficiency and scalability of cloud resource management.

5. Generalization to Other Economic Problems

Similar optimization approaches can be applied in other areas such as healthcare for allocating medical resources, traffic management for reducing congestion, and supply chain management for improving inventory and logistics efficiency.